



Module 3: Technical Core

Electrical Principles · Components · Schematics
2026–2030 Technician Question Pool



Part 1: Electrical Principles

Subelements T5A & T5D · 2 exam questions



The Water Analogy

- Voltage (V) = water pressure — what pushes the current
- Current (I) = flow rate — how much is flowing (measured in amps)
- Resistance (R) = pipe restriction — what opposes the flow (measured in ohms)
- Power (P) = work being done (measured in watts)
- This analogy works for every concept in this section
- High pressure + wide pipe = lots of flow = lots of power



Ohm's Law: $V = I \times R$

- THE formula. Covers 3 of your 35 exam questions by itself.
- Voltage = Current $\times$ Resistance
- Current = Voltage $\div$ Resistance ($I = V/R$)
- Resistance = Voltage $\div$ Current ($R = V/I$)
- Example: 12V battery, 4Ω resistor $\rightarrow I = 12 \div 4 = 3$ amps
- Example: 2A through $50\Omega \rightarrow V = 2 \times 50 = 100$ volts
- Cover the one you want — the other two multiply or divide



Power: $P = V \times I$

- Power in watts = Voltage $\times$ Current
- Example: $13.8V \times 10A = 138$ watts
- Also: $P = I^2 \times R$ (but $V \times I$ is the one on the exam)
- A 100W radio at 13.8V draws about 7.2 amps
- This tells you what size power supply and fuse you need
- Real-world application: that's why your radio comes with a 15A power cord



AC vs. DC

- DC = direct current — flows one direction (batteries, most radio circuits)
- AC = alternating current — flows back and forth (wall outlet = 60 Hz)
- Radio signals are AC — they alternate at the transmit frequency
- Frequency = cycles per second (hertz)
- Period = time for one complete cycle ($1/\text{frequency}$)
- 60 Hz AC = the current reverses direction 120 times per second



Part 2: The Math

Subelement T5B · 1 exam question



Metric Prefixes

- Giga (G) = 1 billion — $1 \text{ GHz} = 1,000 \text{ MHz}$
- Mega (M) = 1 million — $7.2 \text{ MHz} = 7,200,000 \text{ Hz}$
- Kilo (k) = 1 thousand — $1 \text{ k}\Omega = 1,000 \Omega$
- Milli (m) = $1/1,000$ — $500 \text{ mW} = 0.5 \text{ W}$
- Micro (μ) = $1/1,000,000$ — $100 \mu\text{F}$
- Pico (p) = $1/1,000,000,000,000$ — tiny capacitors
- Just move the decimal point — multiply or divide by 1,000 each step



Decibels — Three Facts

- Decibels (dB) measure power ratios, not absolute power
- +3 dB = power DOUBLES
- -3 dB = power HALVES
- +10 dB = power $\times 10$
- That's it. That's all the dB math on the exam.
- Example: 5W + 3 dB = 10W. 5W + 10 dB = 50W.
- Example: 5W + 3 + 3 = 20W (doubled twice)



Part 3: Capacitance & Impedance

Subelement T5C · 1 exam question



Capacitors, Inductors, Impedance

- Capacitor: stores energy in an electric field (water tank)
- Inductor: stores energy in a magnetic field (flywheel)
- Impedance: total opposition to AC current (resistance + reactance)
- Impedance is measured in ohms — just like resistance
- Standard amateur impedance: 50 ohms (everything matches to 50Ω)
- Resonance: when L and C cancel each other → minimum impedance
- Resonant circuits are how antennas and filters are tuned



Part 4: Components

Subelements T6A & T6B · 2 exam questions



Passive Components

Resistors & Capacitors

- Resistor: opposes current flow (zigzag symbol)
- Fixed or variable (potentiometer = volume knob)
- Capacitor: stores energy in electric field (|| symbol)
- Fixed or variable (tuning circuits)
- Fuse: melts to break circuit (overcurrent protection)
- Battery: series = more voltage; parallel = more capacity

Inductors & Semiconductors

- Inductor/choke: stores energy in magnetic field (coil symbol)
- Blocks AC, passes DC — used as filters
- Diode: one-way valve for current (triangle→line)
- LED = diode that emits light
- Transistor: electronic switch/amplifier
- Gain = amplification factor



Part 5: Schematics

Subelements T6C & T6D · 2 exam questions



Schematic Symbol Recognition

- Resistor → zigzag line
- Capacitor → two parallel lines
- Inductor → coiled line
- Diode → triangle pointing to a line
- Transistor → three terminals with arrow
- Battery → long/short parallel lines (long = positive)
- Ground → downward triangle/lines
- Antenna → vertical line with horizontal lines
- Switch → gap in a line
- Know these cold — symbol matching is guaranteed on the exam



Component Functions

- Rectifier — converts AC to DC (made from diodes)
- Voltage regulator — steady output voltage regardless of input
- Relay — small signal controls a large current switch
- Transformer — changes AC voltage (step up or step down)
- Resonant circuit — tuned to specific frequencies (L+C)
- Integrated circuit (IC) — thousands of components in one chip
- These are 'what does X do' exam questions — pure definitions



Key Numbers for the Exam

Formulas & Math

- $V = I \times R$ (Ohm's Law)
- $P = V \times I$ (Power)
- Wavelength = $300 \div \text{MHz}$ (from Module 2)
- +3 dB = double power
- +10 dB = $10\times$ power
- Standard impedance: 50 ohms
- Metric: k=1000, M=1million, m=1/1000

Components

- Capacitor: electric field storage
- Inductor: magnetic field storage
- Diode: one-way current valve
- Transistor: switch/amplifier
- Rectifier: AC $\rightarrow$ DC
- Transformer: changes AC voltage
- LED: diode that emits light



Checkpoint Quiz

10 questions from T5, T6



Checkpoint Quiz

Q1

What formula calculates voltage from current and resistance?

A. $V = I \div R$

B. $V = R \div I$

C. $V = I \times R$

D. $V = I^2 \times R$



Checkpoint Quiz

Q2

What is the current through a 100-ohm resistor connected across 200 volts?

- A. 0.5 amps
- B. 2 amps
- C. 20 amps
- D. 200 amps



Checkpoint Quiz

Q3

How much power is delivered by 13.8 volts at 10 amperes?

- A. 0.72 watts
- B. 23.8 watts
- C. 138 watts
- D. 1380 watts



What does +3 dB represent?

- A. Power is tripled
- B. Power is doubled
- C. Power is halved
- D. Power $\times 10$



What component stores energy in an electric field?

- A. Resistor
- B. Capacitor
- C. Inductor
- D. Transformer



What is the function of a transistor?

- A. Store energy in magnetic field
- B. Convert AC to DC
- C. Switch or amplify signals
- D. Block DC while passing AC



What converts AC to DC?

- A. Capacitor
- B. Inductor
- C. Rectifier
- D. Transformer



Standard impedance of amateur radio systems?

- A. 8 ohms
- B. 50 ohms
- C. 75 ohms
- D. 300 ohms



Checkpoint Quiz

Q9

What is 1,500,000 hertz in megahertz?

- A. 0.15
- B. 1.5
- C. 15
- D. 150



What schematic symbol represents a diode?

- A. Zigzag line
- B. Two parallel lines
- C. Triangle pointing to a line
- D. Coiled line

Before Next Session

Assignment for Module 4



Your Assignment

- Drill HamStudy: T5A–T5D, T6A–T6D to 80%+
- Practice the Ohm's Law calculations until they're automatic
- Coming in Module 4: Signals, Antennas & Safety
 - Station equipment and SWR
 - Digital modes: FT8, APRS, Winlink, DMR
 - Antenna types: dipole, vertical, Yagi
 - Feed lines and connectors
 - Electrical and RF safety
- The hard part is behind you — Module 4 is concrete and visual